Discrete Event Simulation: MBTA's Blue Line Transit Network

Abstract

We model the MBTA Blue Line rapid transit system as a discrete-event simulation, calibrated on real-life trip records (MBTA LAMP portal) from October 2024 and October 2025. Using the Burr distribution to mimic inter-station travel and dwell times, we simulate 6 trains over a 20-hour service period across 100 replications. We find headway deviations (bunching and gaps between trains) caused by delays and unsynchronized schedules, which result from the stochastic nature of travel and dwell times. We quantify inter-train dependence using partial regression.

Why Simulate? Help operators understand: How many trains minimize bunching? What are the optimal dispatch headways to reduce headway deviations? How do train delays propagate through the system?

Background

1. MBTA Blue Line contains 12 stations in total,
    a. Terminal stations: Wonderland (North) and Bowdoin (South)
    b. Railway Yard is in Orient Heights; where trains are dispatched from.
2. Average Cycle time (time taken to loop through a node/station) is $\approx 40\ minutes$
3. Dwell Time: Time spent by train sitting in station
4. Travel Time: Time taken by train to travel from one station to the next
5. Headway: Time difference between 2 trains
6. Vehicle bunching is a well-documented phenomenon, where trains leave uneven spacing/headway due to delays and other schedule irregularities

Model Assumptions

Assumption	Simplicity Gained	Validity Cost	Remark
Infinite queue size for trains waiting to enter station	Avoids modeling signals and varying queue size for stations	Queue size defined by number of signals before station. Queue size rarely exceeds 1, so little validity is lost for this use case	Queue size rarely exceeds 1, so little validity is lost for this use case
No passenger demand modeling	Dwell time treated as exogenous	Real dwell time grows with crowding	-
Terminal segments mirror reverse direction	Fills missing data	Parameters may not be same	Had interpolate missing data. Ignored layover time at terminal station, as data was missing
6 trains, fixed initial dispatch	Matches observed peak service	Ignores operator intervention	Code compatible with initializing different number of trains

Constant 400 seconds headway while dispatching trains	Clean dispatch spacing	Dispatch time varies in practice	Code compatible with initializing different dispatch interval
All train dispatch from Orient Heights towards Wonderland	To maintain equal dispatch headway	Dispatch varies during actual operations	Code compatible with initializing different direction during dispatch

Model Mechanics

- **Entities**: Trains (6), Stations (12), however code is modular to run different configurations
- **Events**:
  - arrival → join platform queue and set departure time (clock + dwell time)
  - departure → sample next travel time, schedule next arrival (clock + travel time)
  - This system follows a G/G/1 queue structure, where arrival times at each station are determined at the moment of departure from station.

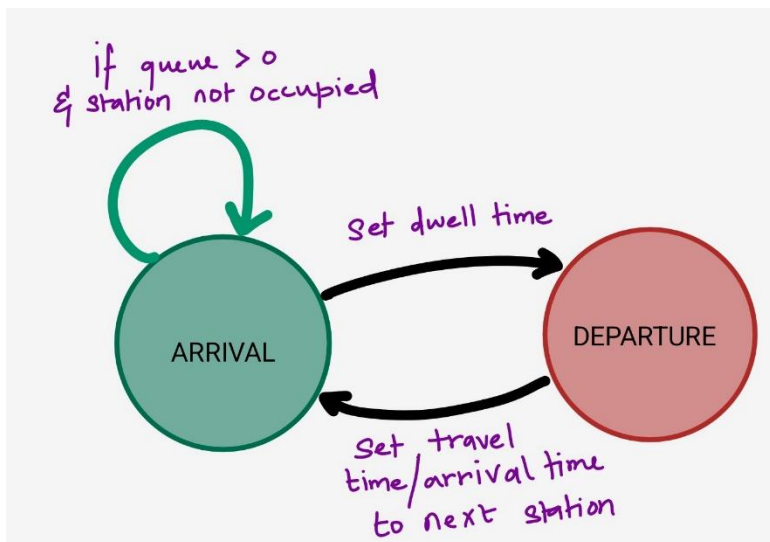


Fig. 1: event decision to simulate train arrival and departure

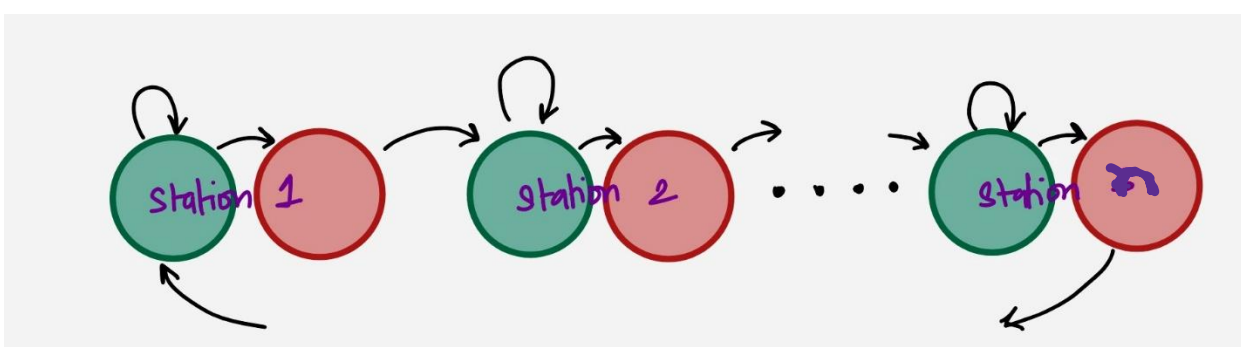


Fig. 2: depicts chained events, arrival time to station generated at previous stop

- **Resources**: Each station-direction pair is a single-server queue (platform). Capacity = 1.

- At time 0, 6 trains are dispatched from Orient Heights at 400-second intervals. Each train arrival event adds the train to the station queue. If the platform is free, the train immediately begins dwelling (sampled from Burr). On departure, travel time to the next station is sampled, the platform is freed, and the next queued train (if any) enters service. Trains reverse direction at Bowdoin and Wonderland (terminal station).
- **Train Attributes**: The following attributes are assigned to each train to track relevant system information for running the simulation and conducting output analysis.
    - train_id: Unique identifier for trains in system
    - station: variable to keep track of dwell or travel station
    - direction: Boolean value specifying trip towards south or north terminal station
    - headway: list of headway time; calculated as difference in departure time in wach station
- **Station Attributes**: Following attributes are assigned to each station to keep track of station level state variables, used during simulation runtime and output analysis.
    - occupied_by: Boolean value to state if station if occupied by a train
    - queue: list of train waiting to enter station; First in – First Out basis
    - last_departure: departure time of previous train to have visited station
    - frequency: time interval between visiting trains

Input Modeling

- **Data**: MBTA LAMP parquet files, Oct 2024 + Oct 2025, filtered to Blue Line (~220K rows after outlier removal)
- **Outlier removal**: Per-station cutoff at $\mu + 3\sigma$ for travel and dwell time. Rationale: removes signal failures and incidents that are not representative of normal operations.
- Dwell and travel times were sampled for each origin-destination station pair
- Distribution selection: Fitted 7 distributions to each of the 20 origin-destination pairs using MLE with floc=0 (time ≥ 0). Compared via AIC:

Distribution	Mean AIC	
	Travel Time	Dwell Time
Weibull	72,017.40	80,766.58
Lognormal	63,929.43	73,745.64
Gamma	64,684.61	74,845.49
Exponential	105,526.74	101,913.37
Burr	59,498.16	71,012.46
Inverse Gaussian	64,056.48	73,879.65
Log Logistic	61,260.12	72,149.99

- A lower AIC score indicates a better fit. The Burr distribution provided the best fit for dwell and travel time distributions among the candidate distributions tested.

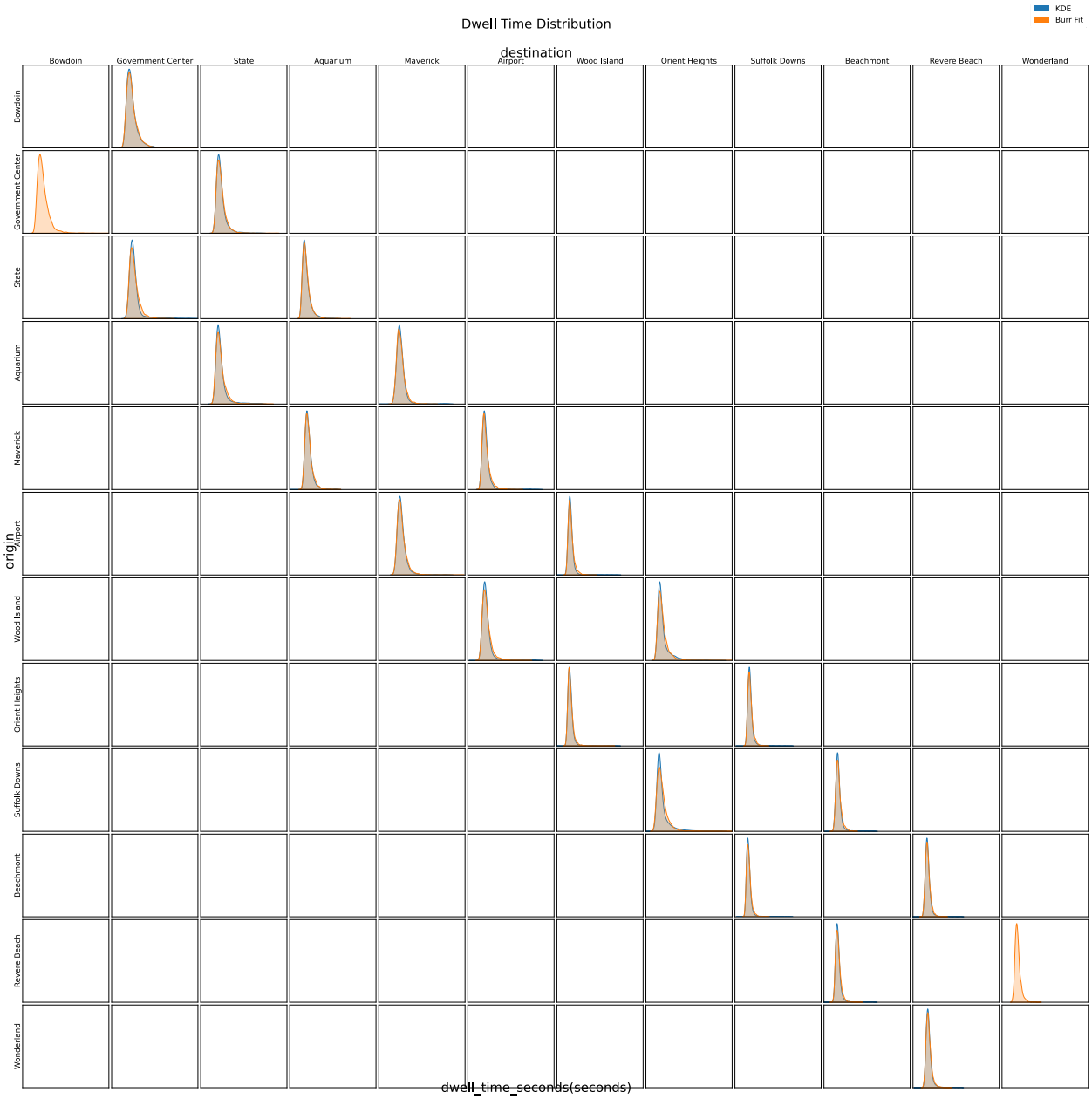


Fig. 3: Fitted Burr distributions for Dwell times across origin-destination station pairs.

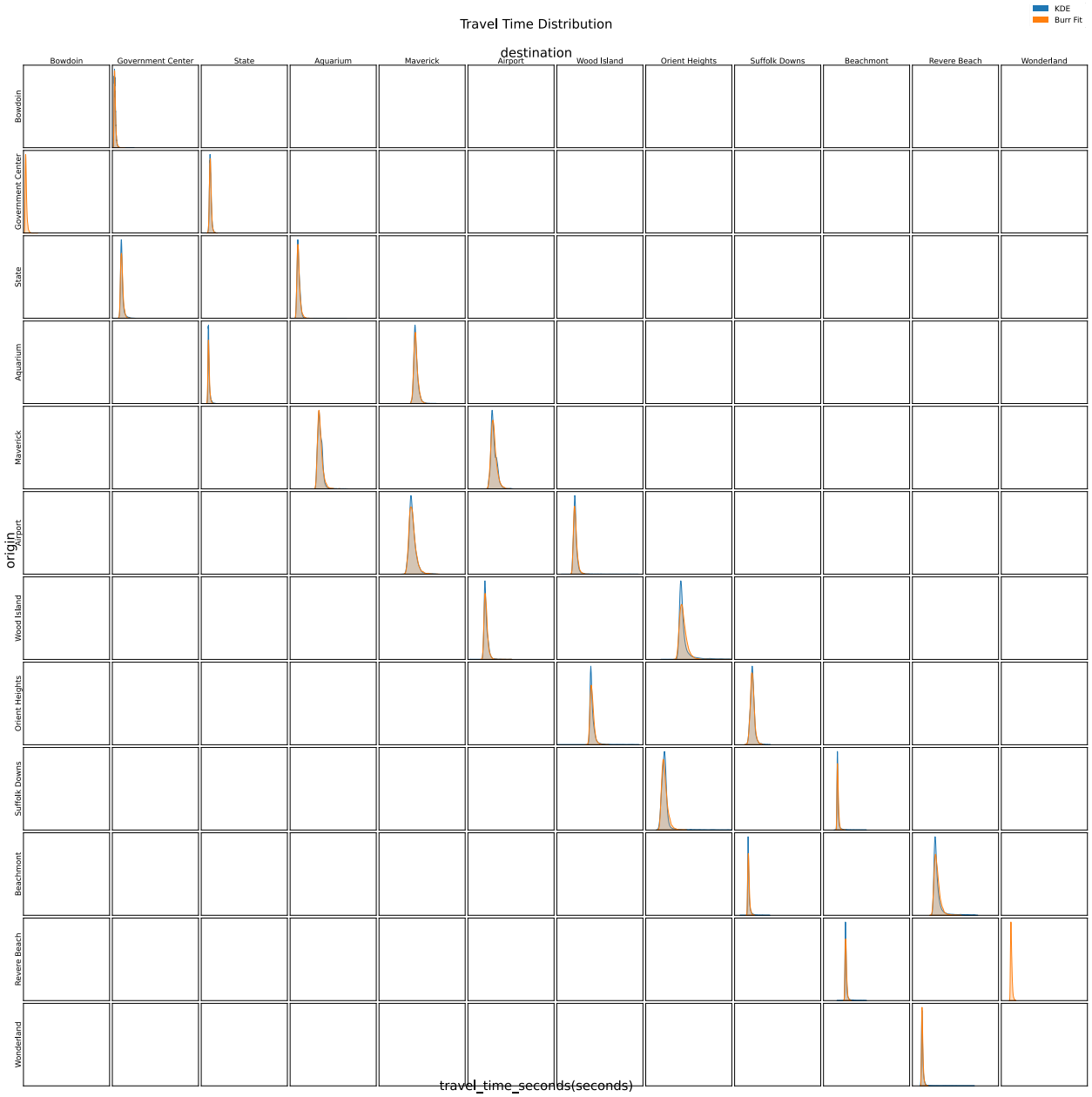


Fig. 4: Fitted Burr distributions for Travel times across origin-destination station pairs.

- The Burr distribution has three parameters:
  - $f(x) = \lambda\left[1 + \left(\frac{x}{\lambda}\right)c\right]d + 1cd\left(\frac{x}{\lambda}\right)c - 1$
  - c (shape)
    - Controls the tail behavior
    - Higher c: lighter tail, distribution decays faster
    - Lower c: heavier tail, more probability in the extremes
  - d (shape)
    - Controls the body/peak shape
    - Higher d: sharper, more concentrated peak
    - Together with c, governs the skewness and kurtosis
  - scale (λ)
    - Stretches or compresses the distribution along the x-axis
    - In your context, directly scales the magnitude of travel/dwell times
    - Higher scale: distribution spread over larger time values

- **Station-to-Station Analysis**
  - Dwell time was examined across individual segments of the line. The analysis revealed that while mean dwell times remain relatively stable across mid-line stations (approximately 30-50 seconds), variance differs substantially from station to station. This distinction is significant: average performance appears consistent, but certain platforms exhibit considerably greater unpredictability in stopping behavior than others.

- **Modelling Rush Hour Conditions**
  - Given the limited data available for peak periods, refitting full distributions for rush hour was not deemed appropriate. Instead, the existing Burr distribution parameters were scaled directly to reflect peak-hour conditions. Within defined time windows (e.g., 9 AM-2 PM), the scale parameter is multiplied by a station-specific scalar, with the parameters *c*, *d*, and *scale* adjusted independently ({'c': 0.928, 'd': 1.131, 'scale': 0.925}). This approach preserves the overall shape of the distribution while stretching its spread to capture the increased variability characteristic of rush hour service.

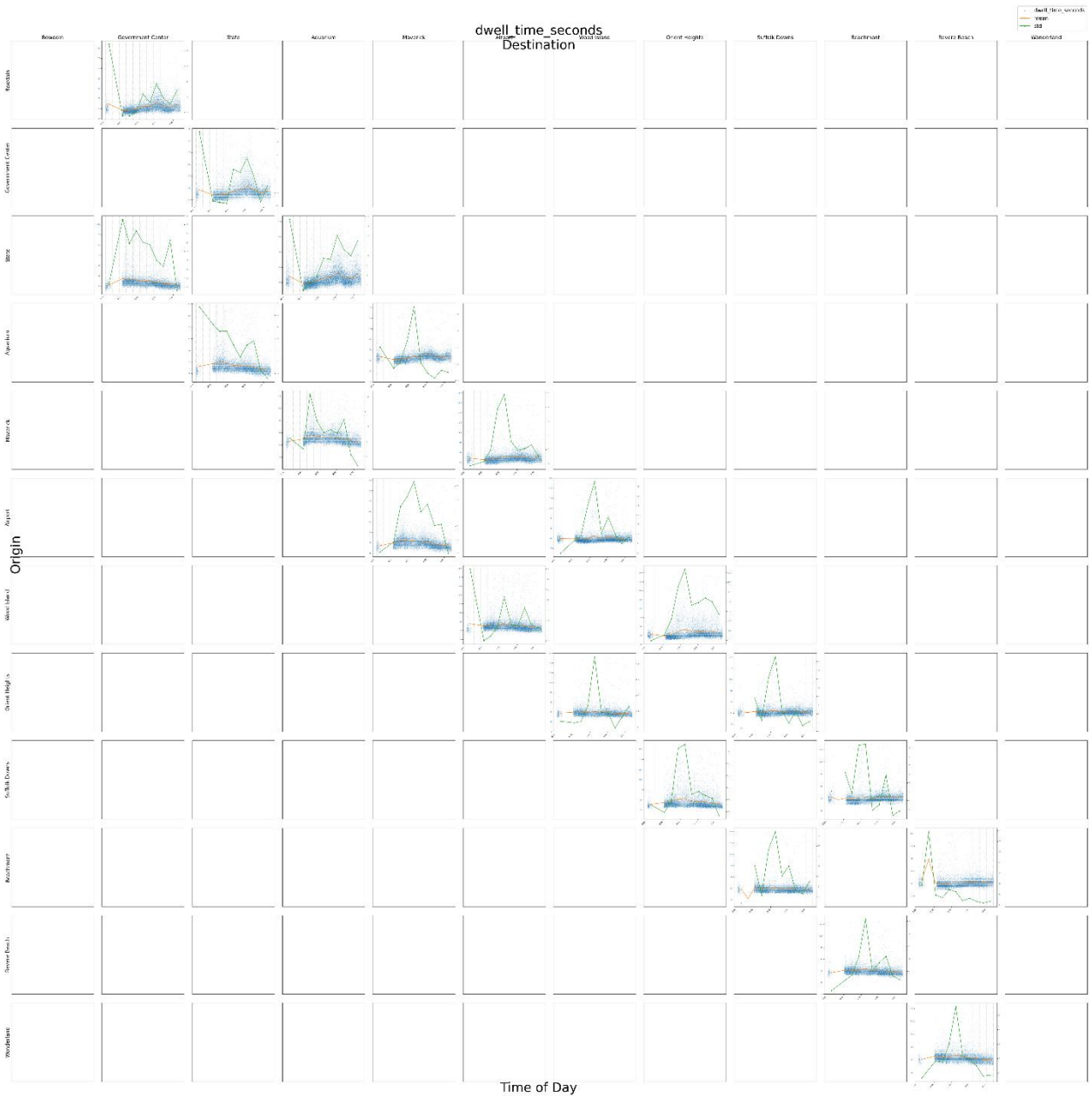


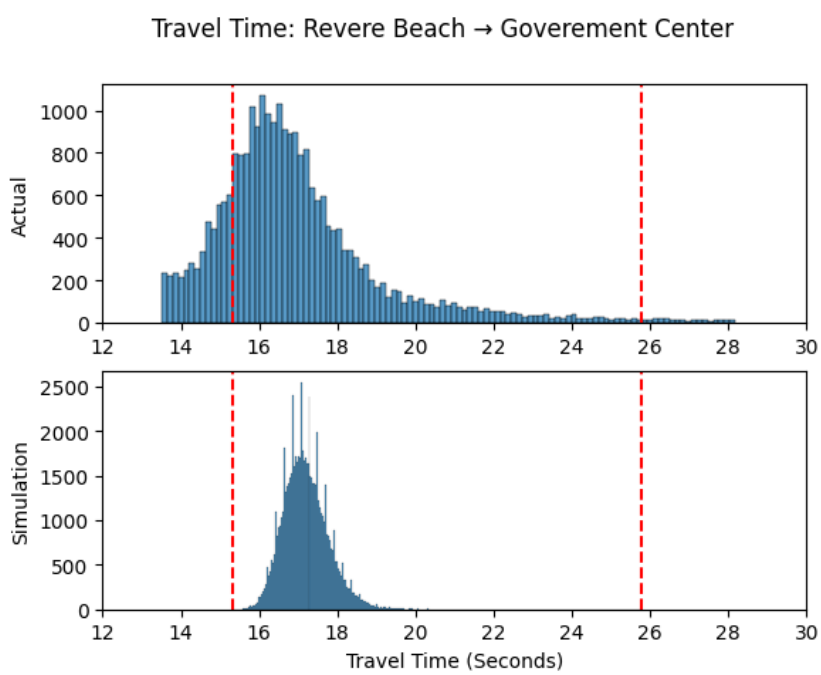
Fig. 5: Mean and Variance of travel time across a day, for each origin-destination station pairs.

Model Validation

- Average cycle time from the official timetable is 39-40 minutes; accurate cycle time could not be determined from the LAMP dataset due to missing terminal station dwell times.
- The table below presents the average cycle time and 10th-90th percentile confidence intervals from the simulation, using different distributions to sample dwell and travel times.

Distribution	Mean	CI(10%)	CI(90%)	Mean AIC	
				Travel Time	Dwell Time
exponential	41.62	33.18	50.78	105,526.74	101,913.37
burr	39.42	38.32	40.58	59,498.16	71,012.46
fisk	38.86	37.83	39.92	61,260.12	72,149.99
lognormal	39.69	38.47	40.97	63,929.43	73,745.64
gamma	39.83	38.51	41.18	64,684.61	74,845.49
weibull	39.27	37.10	41.47	72,017.40	80,766.58

- Simulation travel times spans between 18.24 and 99.01 of actual distribution.



count	25439.000000
mean	16.981268
std	2.232147
min	13.516667
25%	15.633333
50%	16.550000
75%	17.700000
max	28.166667

count	71098.000000
mean	17.194481
std	0.580430
min	15.316667
25%	16.783333
50%	17.150000
75%	17.533333
max	25.783333

Simulation Output Analysis

- Run length: 72,000 seconds (~20 hours); covers a full weekday service window
- Replications: 200
- The table below presents the mean headway time and one standard deviation for each of the six simulated trains in the system.

train_id	Headway	
	Mean	Std
0	397.92	273.56
1	385.99	215.21
2	411.95	209.54
3	391.69	217.35
4	404.82	225.31
5	394.96	230.73

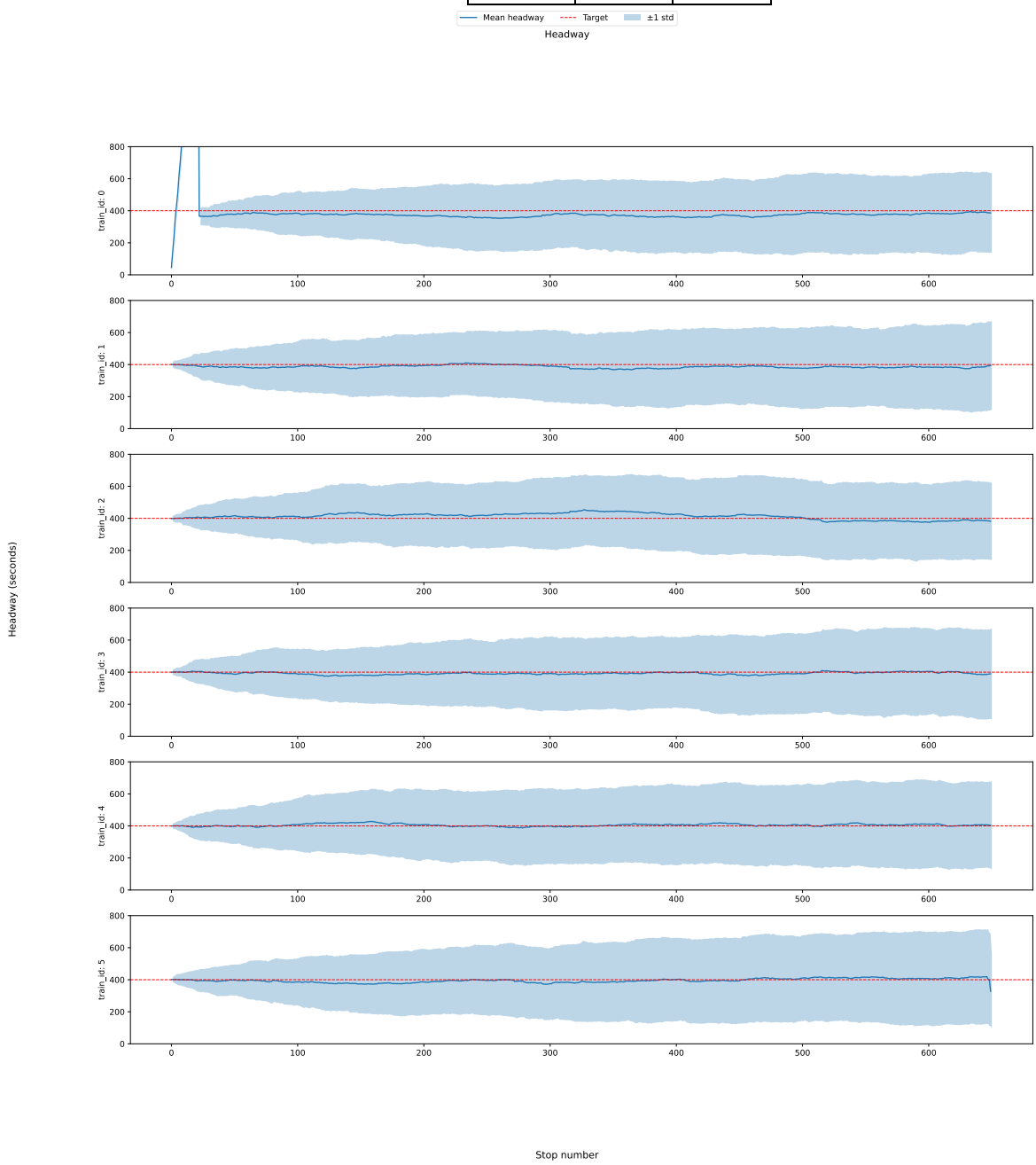


Fig. 6: Average headway time and one standard deviation for each of the six trains, computed over 100 simulation repetitions.

- Trains were initially dispatched at 400-second intervals; however, the resulting average headways deviated from this scheduled spacing due to delays and congestion along the route.
- The following table summarizes the queuing frequency, queue length, and wait time across each station–direction pair, computed over 100 simulation repetitions.

		Frequency		Wait Time		Queue Length
Station	**Direction**	**mean**	**std**	**mean**	**std**	**mean**
Bowdoin	FALSE	346.12	1.53	0.21	0.19	0.91
Government Center	FALSE	340.39	1.52	0.37	0.29	0.94
	TRUE	346.25	1.36	0.29	0.29	0.9
State	FALSE	346.41	1.51	0.25	0.28	0.9
	TRUE	347.51	1.37	0.22	0.21	0.89
Aquarium	FALSE	348.09	1.46	0.15	0.14	0.88
	TRUE	356.21	1.55	0.09	0.11	0.75
Maverick	FALSE	338.03	1.34	0.55	0.4	0.98
	TRUE	344.71	1.44	0.37	0.31	0.94
Airport	FALSE	346.92	1.18	0.3	0.22	0.96
	TRUE	354.74	1.53	0.15	0.18	0.83
Wood Island	FALSE	358.64	1.29	0.08	0.09	0.76
	TRUE	364.74	1.41	0.04	0.06	0.53
Orient Heights	FALSE	351.58	1.44	0.22	0.19	0.87
	TRUE	346.24	1.59	0.22	0.22	0.88
Suffolk Downs	FALSE	356.58	1.33	0.03	0.05	0.54
	TRUE	350.14	1.58	0.09	0.09	0.77
Beachmont	FALSE	355.34	1.29	0.03	0.05	0.59
	TRUE	352.29	1.35	0.04	0.05	0.64
Revere Beach	FALSE	350.87	1.25	0.07	0.07	0.83
	TRUE	352.13	1.3	0.12	0.13	0.77
Wonderland	TRUE	350.49	1.27	0.08	0.08	0.79

- Airport, Government Center, and Maverick were the stations most prone to queuing and congestion, though queue lengths and wait times remained negligible (for system of 6 trains)
- Headway deviation grows monotonically over time. Train 0 (lead train) accumulates the largest deviation; trailing trains show negative correlation when the lead train gaps out, followers bunch up. This is textbook bus bunching, reproduced without any incident injection.
- Within this six-train system, a delay characterized by an increase in headway results in bunching, reflected as a headway decrease, in 75% of cases on average. In the remaining 25% of cases, the relationship between delay and headway change is non-negative.
  - For the first dispatched train, an increase in headway leads to a headway decrease in trailing trains 80% of the time, with the remaining 20% showing no correlation.
  - For all other trains, a headway increases results in trailing trains decreasing their headway in 75% of cases, while the remaining cases exhibit a positive correlation.
- The following figure illustrates the direct effect of delays on trailing trains, as measured through partial regression analysis.

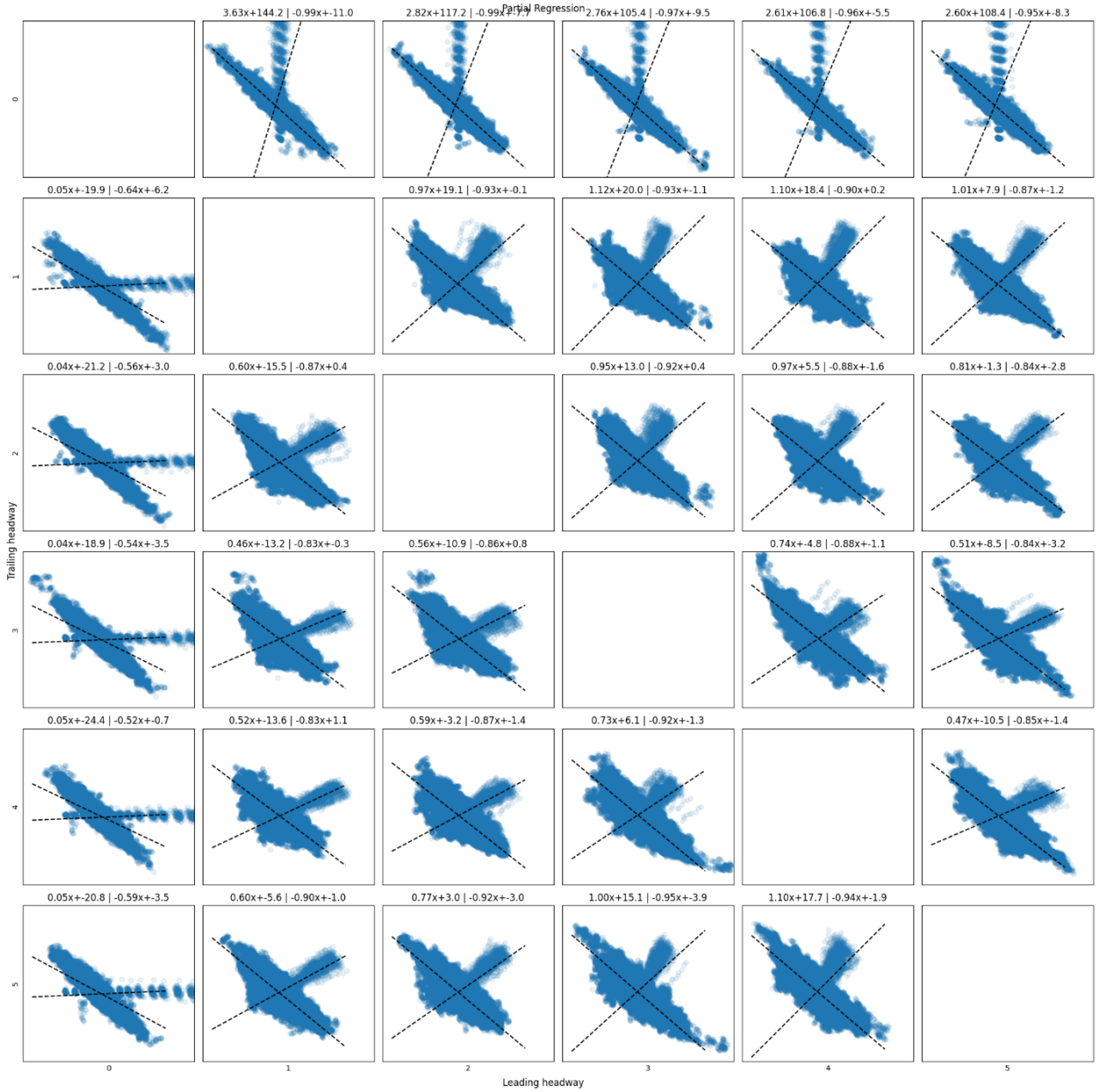


Fig. 7: Partial regression coefficients are estimated separately for each of the six trains.

- Partial regression isolates the effect of one variable on another by statistically controlling for the influence of additional covariates. This allows the direct relationship between a delay and its impact on trailing trains to be estimated independently of other confounding factors, such as dwell time variability or upstream delays.

- As the underlying variables responsible for the negative and non-negative correlations are not observable, the dataset is partitioned by calculating individual pairwise correlations rather than averaging them into a single total correlation. These individual correlations are then used to fit separate regression lines for each case.
  - $cor(individual) = \frac{(x-\widehat{x})(y-\widehat{y})}{\sqrt{(x-\widehat{x})^2(y-\widehat{y})^2}}$

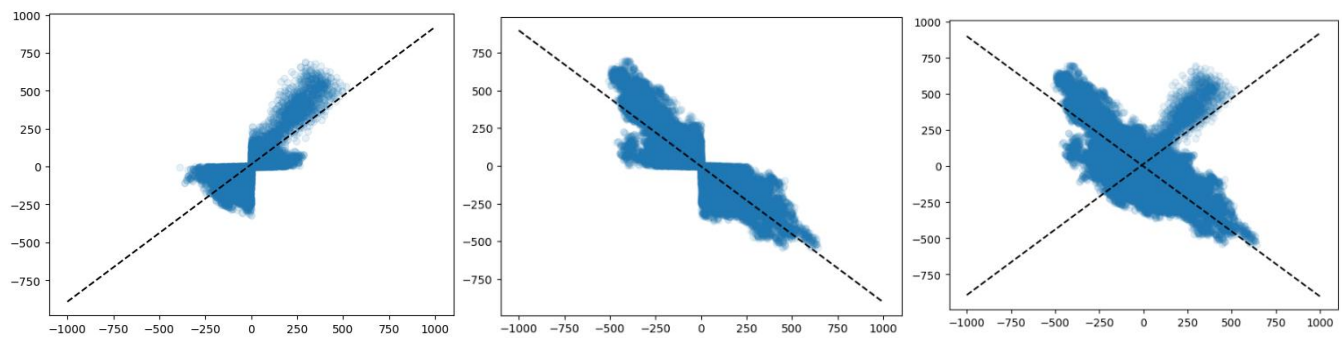


Fig. 8: shows positive and negative trends fitted separately and subsequently combined into a single piecewise relationship.

- Used Nyquist criterion to test whether headway disturbances grow or decay, gain > 1 means bunching amplifies
- Analysis shows:
  - First train is the most vulnerable to instability
  - 4th and 5th train are notorious for causing instability in Train 2
  - 3rd and 4th train are notorious for causing instability in Train 6

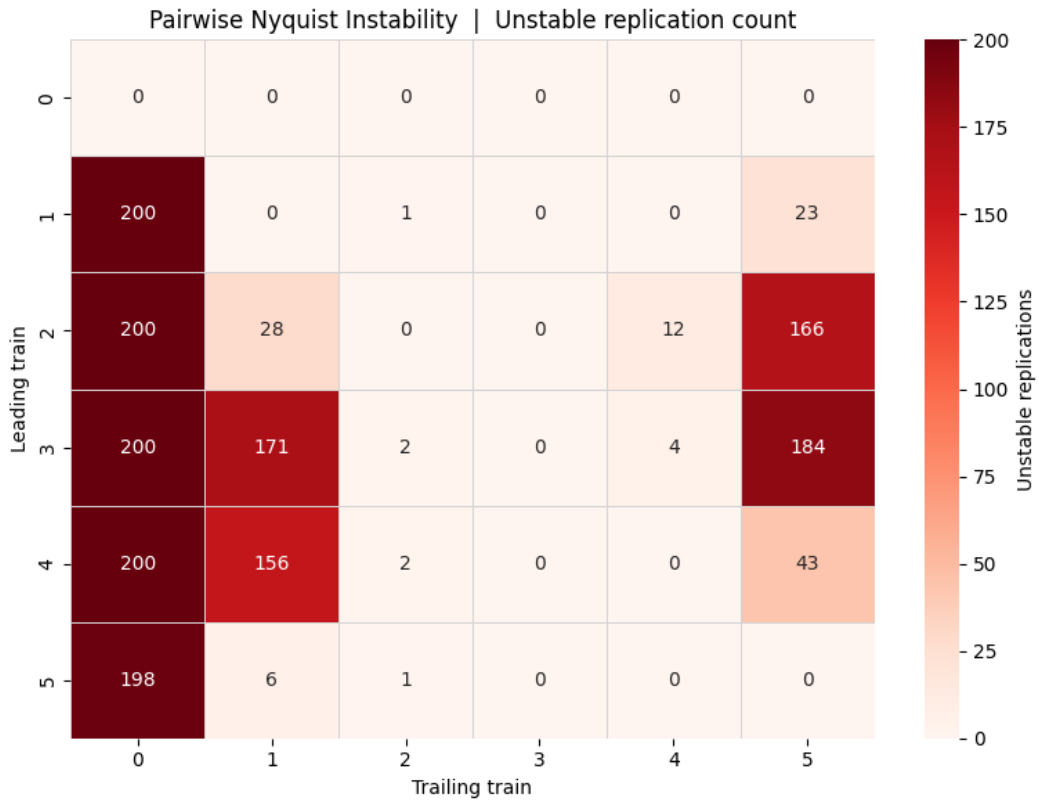


Fig. 9: 6×6 instability heatmap across 200 replications shows risk concentrates in specific upstream downstream train pairs

<u>What if analysis?</u>

1. **Headway Deviation Recovery Analysis**
   - The average headway deviation at the point of dispatch is considered to determine whether such deviations can be closed as trains progress along the line.

train_id	Headway		Dispatch Headway	Deviation
	Mean	Std		
0	397.9	273.56	400	-2.08
1	386	215.21	400	-14.01
2	412	209.54	400	11.95
3	391.7	217.35	400	-8.31
4	404.8	225.31	400	4.82
5	395	230.73	400	-5.04

- Dispatched headway following correction for deviation at the point of dispatch.

Train_id	Headway		Deviation
	mean	std	
0	400.29	289.01	-0.29
1	368.72	211.4	31.28
2	438.13	243.14	-38.13
3	379.15	226.56	20.85
4	391.13	228.93	8.87
5	409.61	230.19	-9.61

- Correcting headway deviation at the point of dispatch alone is not sufficient to eliminate the deviation over the course of the route. This indicates that a real-time control mechanism may be necessary to actively manage and close headway gaps as trains progress along the line.

2. **Impact of Fleet Size on Queue-Induced Delay**
   - Analyze the total delay attributable to queue wait times under varying numbers of trains in the system.

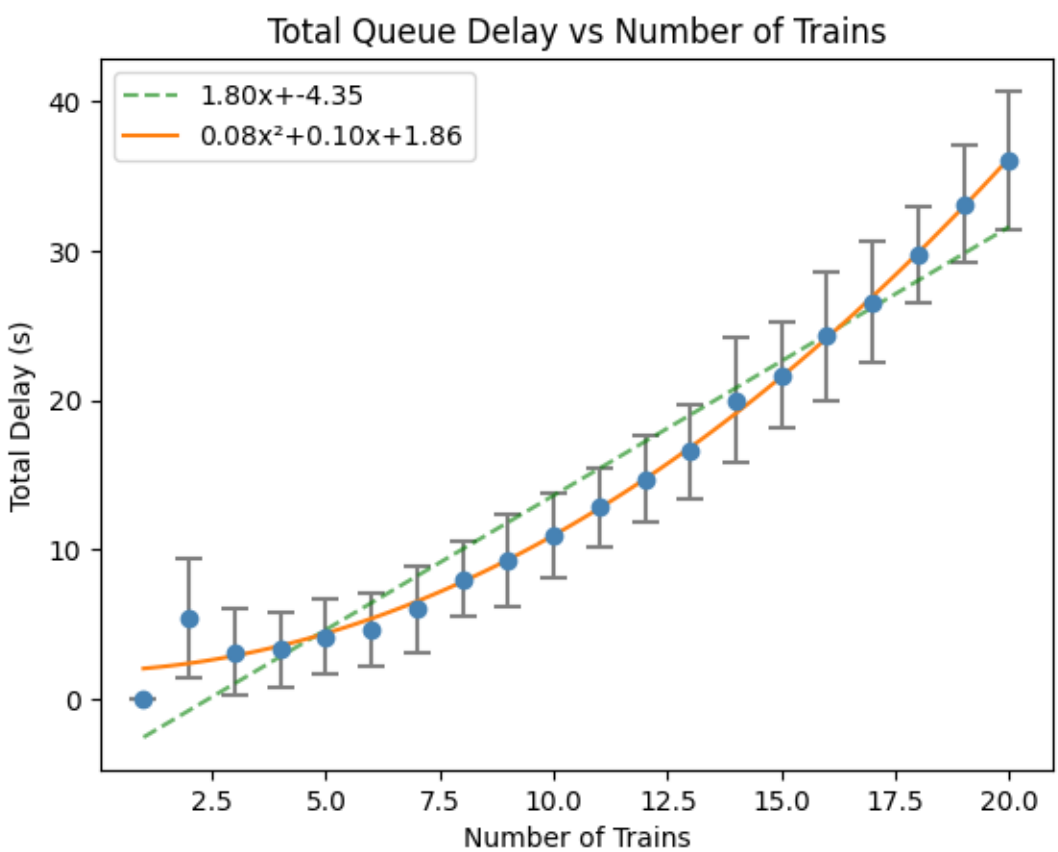


Fig. 10: Linear and polynomial trend lines illustrating the relationship between the number of trains and total delay.

- The data best conforms to a quadratic trend, represented by the equation $0.08x^2 + 0.10x + 1.86$.
- Equivalently best exponential trend is represented by, exp(117.171)

Known Errors & Limitations

- Service parameters for the Government Center → Bowdoin and Revere Beach → Wonderland segments were mirrored from opposing directions, as direct observations were unavailable.
- The absence of layover dwell time distributions at terminal stations represents a key data gap; had this data been available, it could have increased the estimated total cycle time by approximately 9 minutes.
- Partitioning trend lines using correlation does not produce an even spread across the separated data subsets, which may limit how well each trend line represents the true underlying behavior driving the observed patterns.
- The simulation dispatches trains exclusively toward Wonderland, which does not represent real-world operations where trains are distributed across both terminal stations. Consequently, the resulting headway times and service frequency reflect the simulation setup rather than actual operating conditions, potentially limiting the generalizability of the findings.